

Rail Corrugation Problem Write Up

Problem

The Rail Corrugation subsystem requires a file-level classifier that takes a 1-second axle-box sensor recording and predicts one of three conditions: Normal, Side I corrugation, or Side II corrugation. Side I and Side II must be distinguished from the same recording rather than treating corrugation as a single anomaly class.

Data Exploration

The training set contains 272 labelled CSV recordings. Each recording contains 10,000 samples collected at 10,000 Hz across 129 columns: one rotational-speed channel and 128 axle-box channels. The 128 measurement channels correspond to 64 axle boxes, with one vibration and one shock signal for each axle box.

The train contains eight cars with eight axle-box positions per car. Positions 1, 3, 5, and 7 correspond to Side I, while positions 2, 4, 6, and 8 correspond to Side II. This physical mapping was retained throughout the analysis instead of treating all 64 vibration channels as interchangeable.

Class	Files	Share
Normal	234	86.0%
Side I	14	5.1%
Side II	24	8.8%
Total	272	100%

The class distribution is strongly imbalanced, particularly for Side I. Macro F1 was therefore used throughout model comparison because it weights the three class-specific F1 scores equally. Representative Normal, Side I, and Side II files were initially inspected to verify dimensions, channel naming, missing values, and raw signal behaviour. No missing values were found in the inspected recordings.

Signal Analysis

Initial time-domain analysis showed that vibration magnitude contains useful fault information. Across the full training set, mean vibration RMS increased from approximately 0.305 for Normal recordings to 0.580 for Side I and 0.728 for Side II. However, the class distributions overlap substantially, so a simple vibration-amplitude threshold is insufficient.

Class	Overall RMS	Side I RMS	Side II RMS
Normal	0.305	0.303	0.305
Side I	0.580	0.616	0.536
Side II	0.728	0.635	0.803

The side-specific values provided additional localisation information. Side I recordings tended to have stronger vibration on Side I, while Side II recordings showed a clearer increase on Side II. Frequency-domain analysis was then performed using FFT and Welch power spectral density (PSD). A single dominant frequency did not provide sufficient separation, but spectral energy across broader frequency bands showed useful class-dependent differences.

Feature Engineering

Each 1-second recording was converted into a compact fixed-length feature vector. The baseline representation was deliberately side-aware. Time-domain statistics were calculated separately for Side I and

Side II vibration signals, while Welch PSD features captured spectral behaviour over several broad frequency bands.

Feature family	Purpose
Time-domain statistics	RMS, standard deviation, mean absolute value, peak, kurtosis and skew
Side-specific statistics	Preserve differences between Side I and Side II vibration behaviour
Spectral features	Dominant frequency and power across 0-100, 100-250, 250-500, 500-1000, 1000-2000 and 2000-5000 Hz
Side asymmetry	RMS difference and ratio between Side I and Side II

The resulting baseline contained 28 numerical features. Feature quality checks found no NaN or infinite values. Random Forest feature importance indicated that side asymmetry and spectral energy were both useful, with RMS difference, RMS ratio, and several frequency-band power features among the strongest predictors.

Classification and Validation

Because only 14 Side I and 24 Side II recordings are available, a single hold-out split would give a highly unstable estimate. Models were therefore compared using stratified 5-fold cross-validation and Macro F1, matching the competition metric.

Model	Mean Macro F1	Std
Logistic Regression	0.705	0.086
Random Forest	0.716	0.050
Gradient Boosting	0.653	0.089

Random Forest provided the strongest and most stable baseline. Out-of-fold evaluation showed that Normal and Side II were substantially easier to classify than Side I. The baseline class F1 scores were approximately 0.954 for Normal, 0.432 for Side I, and 0.750 for Side II. Side I therefore remained the main limitation of the model.

Feature and Model Experiments

Several controlled experiments were performed to determine whether the baseline could be improved. Alternative PSD aggregation, shock features, explicit train-speed features, wavelength-domain features, feature selection, Random Forest regularisation, explicit class weighting, and Extra Trees were evaluated using the same validation procedure.

Experiment	Mean Macro F1
Baseline vibration features	0.716
Alternative PSD aggregation	0.673
Shock features	0.672
PSD + shock	0.688
Extra Trees	0.634
Speed only	0.386
Baseline + speed	0.681
Wavelength only	0.631

Baseline + wavelength	0.705
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These experiments showed that simply increasing feature count or model complexity did not improve generalisation. Train speed was strongly associated with the class distribution, but speed alone was a poor classifier and adding raw speed reduced performance. A physics-informed wavelength representation was also evaluated by converting vibration frequency to estimated spatial wavelength, but it did not outperform the original frequency-domain representation.

Spatial Axle Analysis

The strongest improvement came from preserving more of the known axle-box geometry. The original baseline compressed all 32 vibration sensors on each side into side-level summaries. Spatial analysis showed systematic differences between corresponding Side I and Side II axle positions, particularly for Side II faults.

Targeted spatial features were therefore added rather than using all 64 axle RMS values directly. These consisted of four paired-position RMS differences, four paired-position RMS ratios, and eight car-level Side I versus Side II RMS differences. The combined spatial representation increased mean stratified 5-fold Macro F1 from approximately 0.716 to 0.731.

Feature set	Mean Macro F1
Baseline	0.716
Pair differences	0.713
Pair differences + ratios	0.717
Car-level differences	0.716
Combined spatial features	0.731

For the combined spatial model, out-of-fold F1 was approximately 0.955 for Normal, 0.438 for Side I, and 0.766 for Side II. The improvement is modest and Side I remains difficult because of the very small number of training examples, so the cross-validation score should be treated as an internal estimate rather than an expected hidden-test score.

Final Model and Test Inference

The final classifier uses the baseline vibration features together with the selected spatial features. A Random Forest with 500 trees, balanced class weighting, and random_state=42 was fitted on all 272 labelled training recordings. The final training matrix contained 44 numerical model features plus filename and label, with no missing or infinite values. The trained classifier was saved as rail_model.pkl.

The same feature-extraction pipeline was then applied to all 68 unlabelled test recordings. The resulting test feature matrix contained no missing or infinite values. The final model produced 58 Normal, 6 Side I, and 4 Side II predictions.

Test inference check	Result
Test recordings	68
Feature NaNs / infinities	0 / 0
Normal predictions	58
Side I predictions	6
Side II predictions	4
Output rows	68

The generated rail_predictions.csv follows the required Rail Corrugation submission schema with one row per test file and the columns file_id and prediction. Predictions are restricted to the exact labels Normal, Side I, and Side II. The final notebook also verifies the expected row count and output schema before saving the submission file.